Modeling, Identification and Control of Mechatronic Systems
FALL 2025

OUT: 09/30/25 DUE: 10/14/25

Use separate sheets to answer each problem with clear problem number, figures, equations and illustrations. Please **submit the answer sheets** together with **MATLAB codes and Simulink model** (MATLAB2022a Version or below) used in your problem analysis when you hand in your homework.

PROBLEM #2 (SYSTEM IDENTIFICATION)

The rotary inverted pendulum system has been described in Problem 1. Derived from the linearized dynamics of the rotary inverted pendulum system, the block diagram of the system is shown in Figure 2.1. Simultaneously, the block diagram of the rotary pendulum system is shown in Figure 2.2. The physical parameters are shown in Table 2.1. Suppose the model parameters (i.e., $\bar{K}_t, \tau_t, K_r, \zeta_r, \omega_r, \bar{K}_a, A_c$) are unknown. Please do system identification with solving the following problems.

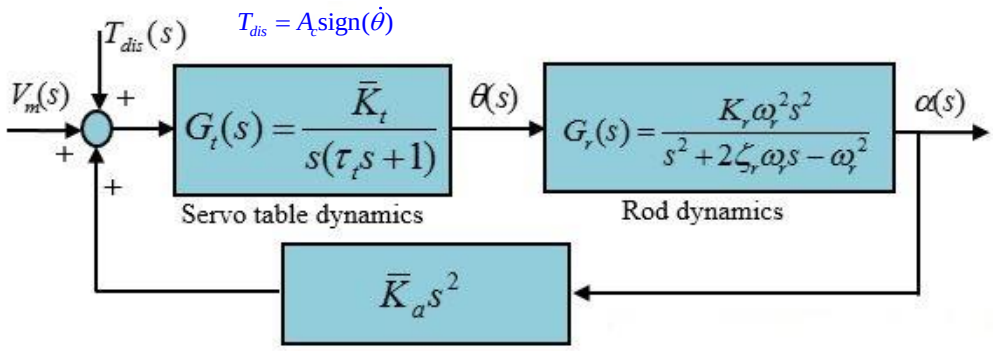

Figure 2.1 Block diagram of the rotary inverted pendulum system

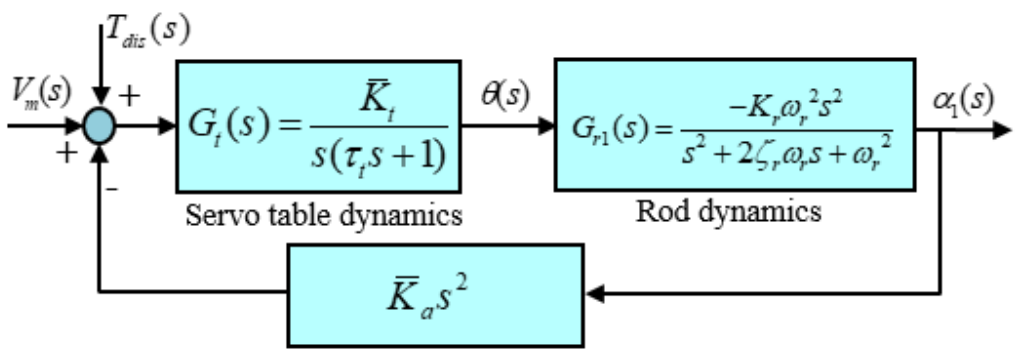

Figure 2.2 Block diagram of the rotary pendulum system

The relationship of model parameters and physical parameters are given by

$$\zeta_r = \sqrt{\frac{B_p^2}{4 m_p L_c I_p g}}, \omega_r = \sqrt{\frac{m_p L_c g}{I_p}}, K_r = \frac{m_p L_c L_t}{I_p \omega_r^2} = \frac{L_t}{g}, \bar{K}_a = \frac{m_p L_t L_c}{K_u}, \tau_t = \frac{I_t}{B_t}, \bar{K}_t = \frac{K_u}{B_t},$$

$$I_p = J_p + m_p L_c^2, I_t = J_t + m_p L_t^2, K_u = \frac{\eta_m \eta_g k_t K_g}{R_m}, B_t = \frac{\eta_g K_g^2 \eta_m k_t k_m + B_r R_m}{R_m}, J_t = J_r$$

Table 2.1 Rotary Pendulum system specifications

Symbol	Description	Value	Unit
m_p	Mass of pendulum	0.127	kg
L_c	Half of the total length of pendulum	0.1685	m
J_p	Pendulum moment of intertia about center of mass	0.0012	kg.m^2
B_p	viscous damping coefficient of the pendulum	0.0024	N.m.s/rad
L_r	Rotary arm length from pivot to tip	0.216	m
J_r	Rotary arm moment of inertia about its center of mass	0.001	kg.m^2
B_r	Rotary arm viscous damping coefficient as seen at the pivot axis	0.0024	N.m.s/rad
R_m	Motor armature resistance	2.6	Ω
k_t	Motor current-torque constant	7.68×10^{-3}	N.m/A
k_m	Motor back-EMF constant	7.68×10^{-3}	V/(rad/s)
K_g	Total gear ratio	70	
η_m	Motor efficiency	0.69	
η_g	Gearbox efficiency	0.9	
A_c	Coulomb friction coefficient of input disturbance	0.4	
V_m	Applied voltage to the motor	-10~10	V

(A) Since the absolute values of model parameters of the rotary inverted pendulum system are the same as that of the rotary pendulum system, we can excite the rotary pendulum system with an appropriate square wave input (for example, a square wave input with frequency of 0.8Hz and amplitude of 0.8V) to do the system identification. According to the linear dynamics of the system, write down the regressor model for system identification, i.e., $y=\varphi^T\vartheta$. Indicate the parameter vector ϑ and its corresponding regressor φ as well as the output y.

(B) Please fill in the necessary MATLAB codes in the attached MATLAB files to complete the system identification. Please describe the relevant techniques with necessary figures and illustrations, such as **generating input signals**, **selecting useful data**, **obtaining parameter estimation**, **verifying results** of system identification etc. The necessary figures should include exciting signal, arm angle, rod angle and their filtered signals, residuals for model validation etc.

Note the filter for obtaining position, velocity and acceleration signal can be

$$H_f(s)=\frac{\omega_f^{\ 2}}{s^2+2\zeta_f\omega_f s+\omega_f^{\ 2}},\ H_{f1}(s)=\frac{\omega_f^{\ 2}s}{s^2+2\zeta_f\omega_f s+\omega_f^{\ 2}},\ H_{f2}(s)=\frac{\omega_f^{\ 2}s^2}{s^2+2\zeta_f\omega_f s+\omega_f^{\ 2}}$$

(C) Compare the system identification results under different cases listed. Comment the effect of these factors (i.e., measurement noise, filter design, input disturbance) on the parameter estimation. What should we notice in the system identification?

	Variance of measurement noise	Identical filter	Input disturbance $d_t=A_d\sin(2\pi f_d t)$
Case 1	$\sigma_n=0$	$\omega_f=15,\zeta_f=0.7$	$A_d=0,f_d=1$
Case 2	$\sigma_n=0$	$\omega_f=5,\zeta_f=0.7$	$A_d=0,f_d=1$
Case 3	$\sigma_n=0$	$\omega_f=300,\zeta_f=0.7$	$A_d=0,f_d=1$
Case 4	$\sigma_n=1$	$\omega_f=15,\zeta_f=0.7$	$A_d=0,f_d=1$

Case 5	$\sigma_n=3$	$\omega_f=15,\zeta_f=0.7$	$A_d=0,f_d=1$
Case 6	$\sigma_n=0$	$\omega_f=15,\zeta_f=0.7$	$A_d=0.1,f_d=1$
Case 7	$\sigma_n=0$	$\omega_f=15,\zeta_f=0.7$	$A_d=0.5,f_d=1$